MIZAN

A Quantitative Framework for Ethical Compliance in Open Source Intelligence Investigations

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Research Draft

github.com/gqnxx/mizan-ethics

Abstract

Modern Open Source Intelligence (OSINT) suffers from two documented failures: the Observer Effect and Adjacent Data Leakage. This paper proposes the MIZAN Protocol, a formal framework designed to quantify and minimize the "Digital Blast Radius" of any investigation through a zero-touch evidence synthesis model and a mathematical heuristic for ethical risk assessment.

1. Introduction

As digital footprints become persistent and interconnected, the line between "public data" and "private life" has effectively dissolved. Social activity, location metadata, and professional connections form a web revealing more than any single point.

Traditional OSINT ethics rely on subjective judgment: "Is it public? Is it justified?" MIZAN Protocol moves from evaluation to **Ethical Attestation**, where every data point carries a provenance tag and a computed Impact Score.

1.1 The Observer Effect

The act of investigating a identity leaves traces. Platform notification systems (LinkedIn profile views, Instagram receipts) alert the target, causing behavioral changes or evidence destruction.

1.2 Adjacent Data Leakage

Non-targets (family, colleagues) are inadvertently exposed. In some regions, family networks are denser, meaning a single search can expose an entire social structure.

2. Blast Radius Calculus (BRC)

The core contribution is the Blast Radius Calculus, a heuristic for determining ethical risk score (ERS) before execution.

$$ERS = \Phi(S, A, E)$$

- **S (Sensitivity):** Privacy value of the data (0.0 to 1.0).
- **A (Adjacency):** Degree of separation from target.
- **E (Entropy):** Disturbance caused by the method.

2.1 Threshold Logic

A step is approved only if $ERS < \tau$, where τ is the Ethical Threshold defined by context (Law Enforcement, Corporate, etc.).

3. Minimal-Interaction Evidence Synthesis (MIES)

Methodology for collection with zero touch:

- **Cache-First:** Prioritize cached content (Wayback, Google Cache).
- **Proxy Synthesis:** Route through layers that eliminate fingerprints.
- **Temporal Separation:** Avoid related data access in narrow windows.
- **Zero-Footprint:** Test methods against control environments.

4. Proof-of-Ethical-Conduct (PoEC)

Every piece of evidence must carry a PoEC signature, creating a digital chain of custody:

- **Source Pathway:** URL or API used.
- **Access Method:** Tool and configuration used.
- **ERS Score:** Calculated risk at time of collection.
- **Logic:** Justification for non-consensual access.
- **Integrity Hash:** SHA-256 hash of the evidence.

5. Conclusion

MIZAN is a security architecture for future digital investigators. By quantifying ethics through BRC, enforcing stealth through MIES, and documenting compliance through PoEC, we protect both the subject and the investigator.